

SF2940 Probability: Exam

Fall 2025

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Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of four independent problems. Each problem gives at most 5 points.

Problem 1. Let X and Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} \frac{4}{x^3 y}, & 1 < y < x, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine $\mathbb{E}[Y | X]$ (2 points)
(b) Let

$$U = \frac{X}{X+Y} \quad \text{and} \quad V = X+Y.$$

Determine $f_{U,V}(u,v)$. Are U and V independent? Thoroughly motivate your answer. (3 points)

Solution. (a) Recall that since X and Y are continuous,

$$\mathbb{E}[Y | X = x] = \int_{-\infty}^{\infty} y f_{Y|X=x}(y) dy,$$

where

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

Thus, we need to compute $f_X(x)$:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_1^x \frac{4}{x^3 y} dy = \frac{4}{x^3} \int_1^x \frac{1}{y} dy = \frac{4}{x^3} [\log y]_1^x = \frac{4 \log x}{x^3}$$

Consequently,

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{4}{x^3 y}}{\frac{4 \log x}{x^3}} = \frac{1}{y \log x} \quad \text{for } 1 < y < x,$$

and $f_{Y|X=x}(y) = 0$ otherwise. Thus, we have that

$$\mathbb{E}[Y | X = x] = \int_{-\infty}^{\infty} y f_{Y|X=x}(y) dy = \int_1^x y \frac{1}{y \log x} dy = \frac{1}{\log x} \int_1^x dy = \frac{x-1}{\log x}.$$

Since $\mathbb{E}[Y | X = x] = \frac{x-1}{\log x}$, we have that

$$\mathbb{E}[Y | X] = \frac{X-1}{\log X}.$$

(b) We let

$$U = \frac{X}{X+Y} \quad \text{and} \quad V = X+Y.$$

Then,

$$\begin{aligned} X &= UV =: h_1(U, V) \\ Y &= V - X = V - UV =: h_2(U, V). \end{aligned}$$

By the transformation theorem,

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) |\mathbf{J}|,$$

where

$$\mathbf{J} = \begin{vmatrix} \frac{\partial}{\partial u} h_1(u, v) & \frac{\partial}{\partial v} h_1(u, v) \\ \frac{\partial}{\partial u} h_2(u, v) & \frac{\partial}{\partial v} h_2(u, v) \end{vmatrix} = \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix} = v(1-u) - (-v)u = v.$$

Consequently,

$$\begin{aligned} f_{U,V}(u, v) &= f_{X,Y}(uv, v(1-u)) |v| \\ &= \frac{4}{(uv)^3 \cdot v(1-u)} |v| \\ &= \frac{4}{u^3(1-u)v^4} |v|, \quad \text{when } 1 < v(1-u) < uv \end{aligned}$$

and $f_{U,V}(u, v) = 0$ otherwise.

Now, we quantify the bounds $1 < v(1-u) < uv$. Note first that since $uv > 1 > 0$, we must have that u and v have the same sign (either both positive or both negative). However, since

$$1 < v(1-u) \iff 0 < 1+uv < v,$$

we have $u > 0$ and $v > 0$. In particular, it means that in this range of parameters, $|v| = v$, so that

$$f_{U,V}(u, v) = \frac{4}{u^3(1-u)v^4} |v| = \frac{4}{u^3(1-u)v^3}.$$

Next, we note that

$$v(1-u) < uv \iff 1-u < u \iff 1 < 2u \iff u > \frac{1}{2}$$

Moreover, since $v > 0$ and $v(1-u) > 1 > 0$, we must have $u < 1$. Finally, since $\frac{1}{2} < u < 1$ (in particular, so that $1-u > 0$), we have that

$$1 < v(1-u) \iff v > \frac{1}{1-u}.$$

Thus, the bounds $1 < v(1 - u) < uv$ can be quantified as

$$\frac{1}{2} < u < 1, \quad v > \frac{1}{1-u}$$

Thus,

$$f_{U,V}(u, v) = \begin{cases} \frac{4}{u^3(1-u)v^3} & \text{if } \frac{1}{2} < u < 1, \quad v > \frac{1}{1-u}, \\ 0, & \text{otherwise.} \end{cases}$$

Finally, we investigate the if U and V are independent or not. We note that

$$f_U(u) = \int_{-\infty}^{\infty} f_{U,V}(u, v) dv = \int_{\frac{1}{1-u}}^{\infty} \frac{4}{u^3(1-u)} v^{-3} dv = \frac{4}{u^3(1-u)} \left[-\frac{v^{-2}}{2} \right]_{\frac{1}{1-u}}^{\infty} = \frac{2(1-u)}{u^3}.$$

Thus, we note that if U and V are independent, then

$$f_{U,V}(u, v) = f_U(u)f_V(v),$$

that is, for $\frac{1}{2} < u < 1, \quad v > \frac{1}{1-u},$

$$f_V(v) = \frac{f_{U,V}(u, v)}{f_U(u)} = 2 \frac{1}{(1-u)^2 v^3},$$

which cannot be true, since the expression on the right-hand side depends on u . Thus, U and V are not independent.

Problem 2. Let $\mathbf{X} = (X_1, X_2)'$ be a bivariate random vector whose joint probability density is given by

$$f_{X_1, X_2}(x_1, x_2) = C \exp\left(-\frac{3}{4}x_1^2 + x_1x_2 - \frac{1}{2}x_2^2\right), \quad x_1, x_2 \in \mathbb{R},$$

where C is a normalization constant.

- (a) Show that $\mathbf{X}$ has a bivariate (two-dimensional) normal distribution and determine its mean vector and covariance matrix.....(3 points)
- (b) Define the random variable, $Y := X_1 \mathbf{1}_{\{|X_1| \leq 1\}} - X_1 \mathbf{1}_{\{|X_1| > 1\}}$. Show that Y is a normal random variable, but $(X_1, Y)'$ is not a bivariate (two-dimensional) normal.(2 points)

(Recall that for an event A , $\mathbf{1}_A = \mathbf{1}_A(\omega)$ is the random variable which takes the value 1 if $\omega \in A$ and 0 otherwise.)

Solution. (a) We have that $(X_1, X_2)'$ is a bivariate normal random vector if it has joint probability density

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} \frac{1}{\sqrt{\det(\Lambda)}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \Lambda^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} = (x_1, x_2)' \in \mathbb{R}^2.$$

In this case, $(X_1, X_2)' \sim N(\boldsymbol{\mu}, \Lambda)$. Thus, we shall show that we can find $\boldsymbol{\mu} = (x_1, x_2)' \in \mathbb{R}^2$ and a 2×2 -matrix Λ , such that

$$-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \Lambda^{-1}(\mathbf{x} - \boldsymbol{\mu}) = -\frac{3}{4}x_1^2 + x_1x_2 - \frac{1}{2}x_2^2$$

Since the right-hand side has no linear terms, we must have that $\boldsymbol{\mu} = 0$, so that

$$-\frac{1}{2}\mathbf{x}' \Lambda^{-1} \mathbf{x} = -\frac{3}{4}x_1^2 + x_1x_2 - \frac{1}{2}x_2^2 \quad (1)$$

Assuming that $\Lambda^{-1} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ (since Λ is symmetric, so is Λ^{-1}), we have that

$$-\frac{1}{2}\mathbf{x}' \Lambda^{-1} \mathbf{x} = -\frac{1}{2}(ax_1^2 + 2bx_1x_2 + cx_2^2).$$

Consequently, (1) holds if

$$a = \frac{3}{2}, \quad b = -1, \quad c = 1.$$

Thus, $(X_1, X_2)'$ is a bivariate normal random vector and its mean vector is $\boldsymbol{\mu} = (0, 0)'$ and its covariance matrix Λ , which is the inverse of $\begin{pmatrix} \frac{3}{2} & -1 \\ -1 & 1 \end{pmatrix}$. We compute Λ :

$$\Lambda = \frac{1}{\det(\Lambda^{-1})} \begin{pmatrix} 1 & 1 \\ 1 & \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$$

Thus, $(X_1, X_2)' \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}\right)$.

(b) Note that $X_1 \sim N(0, 2)$. We shall first show that Y is a normal random variable, in particular, that $Y \stackrel{d}{=} X_1$. We note that if $y < -1$, then

$$Y \leq y \iff X_1 \geq -y,$$

so that

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X_1 \geq -y) = \mathbb{P}(X_1 \leq y) = F_{X_1}(y),$$

where we used that $X_1 \stackrel{d}{=} -X_1$, in the last equality. Similarly, if $y > 1$, we have that

$$Y > y \iff X_1 < -y,$$

so that

$$1 - F_Y(y) = \mathbb{P}(Y > y) = \mathbb{P}(X_1 < -y) = \mathbb{P}(X_1 > y) = 1 - F_{X_1}(y),$$

so that $F_Y(y) = F_{X_1}(y)$. In particular, since distribution functions are right-continuous, we also have that $F_Y(1) = \lim_{y \searrow 1} F_Y(y) = \lim_{y \searrow 1} F_{X_1}(y) = F_{X_1}(1)$. Finally, if $y \in [-1, 1]$, we have that

$$Y \in (y, 1] \iff X_1 \in (y, 1],$$

so that

$$F_Y(1) - F_Y(y) = F_{X_1}(1) - F_{X_1}(y).$$

Since $F_Y(1) = F_{X_1}(1)$, this means that $F_Y(y) = F_{X_1}(y)$.

Thus, we have proved that for all $y \in \mathbb{R}$,

$$F_Y(y) = F_{X_1}(y),$$

that is, $Y \stackrel{d}{=} X_1$ and hence Y is normal.

We shall now show that $(Y, X_1)'$ is **not** a bivariate normal random vector. We have that $(Y, X_1)'$ is a bivariate normal vector if and only if for each $a, b \in \mathbb{R}$, there exist $\mu, \sigma^2 \in \mathbb{R}$ such that

$$aY + bX_1 \sim N(\mu, \sigma^2).$$

In particular, we must have that $X_1 + Y$ is a one-dimensional normal random variable. However,

$$\begin{aligned} X_1 + Y &= X_1 + X_1 \mathbf{1}_{\{|X_1| \leq 1\}} - X_1 \mathbf{1}_{\{|X_1| > 1\}} \\ &= X_1 \mathbf{1}_{\{|X_1| \leq 1\}} + X_1 \mathbf{1}_{\{|X_1| > 1\}} + X_1 \mathbf{1}_{\{|X_1| \leq 1\}} - X_1 \mathbf{1}_{\{|X_1| > 1\}} \\ &= 2X_1 \mathbf{1}_{\{|X_1| \leq 1\}}, \end{aligned}$$

which has bounded support. Indeed, $\mathbb{P}(|X_1 + Y| \leq 2) = 1$, which cannot be true for any normal random variable. Hence, $(Y, X_1)'$ is not a bivariate normal vector.

Problem 3.

- (a) Recall that a random variable X is said to be symmetric if $X \stackrel{d}{=} -X$. Prove that if X is symmetric, then $\varphi_X(t)$ is real for all $t \in \mathbb{R}$ (1 point)
- (b) Let X and Y be independent and identically distributed random variables. Show that $X - Y$ is symmetric.....(1 point)
(Hint: You may use the fact that if Z is a random variable and $\varphi_Z(t)$ is real for all t , then Z is symmetric.)
- (c) Prove that one cannot find independent and identically distributed random variables X and Y such that

$$X - Y \sim U(-1, 1).$$

..... (2 points)

- (d) Let X and Y be independent and identically distributed random variables and assume that $X - Y \sim \text{Tri}(-1, 1)$. What is the distribution of X and Y ? (1 point)
(Hint: $\varphi_{\text{Tri}(-1,1)}(t) = \left(\frac{\sin(t/2)}{t/2}\right)^2$.)

Solution. (a) Since $X \stackrel{d}{=} -X$, we have $\varphi_X(t) = \varphi_{-X}(t)$ for all $t \in \mathbb{R}$. Moreover, by standard properties of characteristic functions, we also have

$$\varphi_{-X}(t) = \varphi_X(-t)$$

(since $\varphi_{aX+b}(t) = e^{itb}\varphi_X(at)$ for $a, b \in \mathbb{R}$) and

$$\varphi_X(-t) = \overline{\varphi_X(t)}.$$

Putting the above identities together, we have that

$$\varphi_X(t) = \varphi_{-X}(t) = \varphi_X(-t) = \overline{\varphi_X(t)},$$

that is, $\varphi_X(t)$ is real.

(b) Since X and Y are i.i.d., we have that

$$\varphi_{X-Y}(t) = \varphi_X(t)\varphi_{-Y}(t) = \varphi_X(t)\varphi_Y(-t) = \varphi_X(t)\varphi_X(-t) = \varphi_X(t)\overline{\varphi_X(t)} = |\varphi_X(t)|^2,$$

where, in the first equality, we use the independence of X and Y ; in the second equality, we use the identity $\varphi_{aY+b}(t) = e^{itb}\varphi_Y(at)$; in the third equality, we use $X \stackrel{d}{=} Y$; in the fourth equality, we use that $\varphi_X(-t) = \overline{\varphi_X(t)}$; and in the fifth equality, that $z \cdot \bar{z} = |z|^2$ for each complex number z .

In particular, the above shows that $\varphi_{X-Y}(t)$ is real, so that by the hint, $X - Y$ is a symmetric random variable.

(c) We have from the table of distributions, that if $Z \in U(-1, 1)$, then

$$\varphi_Z(t) = \frac{e^{it} - e^{-it}}{2ti} = \frac{\sin(t)}{t}.$$

This characteristic function is real, and does take negative values for some $t \in \mathbb{R}$, since $\sin(t)$ does. Meanwhile, we see in the calculation carried out in part (b) that $\varphi_{X-Y}(t)$ is nonnegative whenever X and Y are i.i.d. and hence we cannot have that $X - Y \sim U(-1, 1)$.

(d) By uniqueness of characteristic functions, $\varphi_{X-Y}(t) = \varphi_{\text{Tri}(-1,1)}(t)$ for all $t \in \mathbb{R}$. Since by the calculation in (b), we have that $\varphi_{X-Y}(t) = |\varphi_X(t)|^2$, and by the hint

$$\varphi_{\text{Tri}(-1,1)}(t) = \left(\frac{\sin(t/2)}{t/2} \right)^2.$$

Thus, we must have that (see the table of distributions)

$$\varphi_X(t) = \frac{\sin(t/2)}{t/2} = \frac{e^{it/2} - e^{-it/2}}{it} = \varphi_{U(-1/2,1/2)}(t).$$

Thus X and Y have distribution $U(-1/2, 1/2)$.

Problem 4. Let $X_1, X_2, \dots$ be independent $\text{Exp}(1)$ -distributed random variables and set $Z_n = \max_{1 \leq k \leq n} X_k$.

- (a) Show that $F_{Z_n}(z) = (F_{X_1}(z))^n$ (1 point)
- (b) Show that $\frac{Z_n}{\log n} \xrightarrow{p} 1$ as $n \rightarrow \infty$ (2 points)
- (c) Let $N \sim \text{Po}(m)$, $m > 0$, be independent of $X_1, X_2, \dots$ and define $Z_0 := 0$. Let $Y \sim \text{Exp}(1/m)$. Show that e^{-Z_N} has the same distribution as $\min(1, Y)$. (2 points)
(Hint: It is useful to first find the distribution function F_{Z_N} .)

Potentially useful identity: $g_{\text{Po}(m)}(t) = e^{m(t-1)}$

Solution. (a) By definition,

$$Z_n = \max_{1 \leq k \leq n} X_k \leq z \iff X_k \leq z \text{ for all } k = 1, 2, \dots, n.$$

Thus, we have that

$$\begin{aligned} F_{Z_n}(z) &= \mathbb{P}(Z_n \leq z) = \mathbb{P}(X_1 \leq z, \dots, X_n \leq z) \\ &= \mathbb{P}(X_1 \leq z) \cdots \mathbb{P}(X_n \leq z) = (\mathbb{P}(X_1 \leq z))^n = (F_{X_1}(z))^n, \end{aligned}$$

where we used the independence of $X_1, X_2, \dots$ in the third equality and that they are identically distributed in the fourth equality.

(b) Note that since $X_1 \sim \text{Exp}(1)$, we have that $F_{X_1}(z) = 1 - e^{-z}$. Thus,

$$\begin{aligned} F_{\frac{Z_n}{\log n}}(z) &= \mathbb{P}\left(\frac{Z_n}{\log n} \leq z\right) = \mathbb{P}(Z_n \leq z \log n) = F_{Z_n}(z \log n) = (F_{X_1}(z \log n))^n \\ &= (1 - e^{-z \log n})^n = \left(1 - e^{\log(\frac{1}{n^z})}\right)^n = \left(1 - \frac{1}{n^z}\right)^n. \end{aligned}$$

Since $(1 - 1/n)^n \rightarrow e^{-1}$ as $n \rightarrow \infty$, we have that

$$F_{\frac{Z_n}{\log n}}(z) = \left(1 - \frac{1}{n^z}\right)^n \rightarrow \begin{cases} 0 & \text{if } z < 1, \\ e^{-1} & \text{if } z = 1 \\ 1 & \text{if } z > 1. \end{cases}$$

Thus, $F_{\frac{Z_n}{\log n}}(z)$ converges to the distribution function F_1 of the constant 1 at every point except for the discontinuity point $z = 1$ (recall that $F_1(z) = 0$ if $z < 1$ and $F_1(z) = 1$ if $z \geq 1$). Thus, we have that $\frac{Z_n}{\log n} \xrightarrow{d} 1$ and since the limit is constant, we have that

$$\frac{Z_n}{\log n} \xrightarrow{p} 1$$

since convergence in distribution to a constant is equivalent to convergence in probability to a constant (see Theorem 3.2 of Chapter 6 in Gut's book).

(c) Let $W = \min(1, Y)$. We begin by determining $F_W(w)$. We note that if $w \geq 1$, then clearly $F_W(w) = 1$. Similarly, since $\mathbb{P}(Y > 0) = 1$, $F_W(w) = 0$, whenever $w \leq 0$. Finally, if $w \in (0, 1)$, we have that,

$$F_W(w) = \mathbb{P}(W \leq w) = \mathbb{P}(Y \leq w) = 1 - e^{-mw}.$$

Thus,

$$F_W(w) = \begin{cases} 0 & \text{if } w \leq 0, \\ 1 - e^{-mw} & \text{if } 0 < w < 1, \\ 1 & \text{if } w \geq 1. \end{cases}$$

As suggested in the hint, we first compute F_{Z_N} . The case $z < 0$ is trivial since $\mathbb{P}(Z_N \geq 0) = 1$. Moreover, $F_{Z_N}(0) = \mathbb{P}(Z_N = 0) = \mathbb{P}(N = 0) = e^{-m}$. Consider $z > 0$, and note that, by the law of total probability,

$$\begin{aligned} F_{Z_N}(z) &= \mathbb{P}(Z_N \leq z) = \sum_{n=0}^{\infty} \mathbb{P}(Z_N \leq z | N = n) \mathbb{P}(N = n) && \text{(law of total probability)} \\ &= \sum_{n=0}^{\infty} \mathbb{P}(Z_n \leq z | N = n) \mathbb{P}(N = n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}(Z_n \leq z) \mathbb{P}(N = n) && (N \text{ independent of } X_1, X_2, \dots) \\ &= \sum_{n=0}^{\infty} F_{Z_n}(z) \mathbb{P}(N = n) \\ &= \sum_{n=0}^{\infty} (F_{X_1}(z))^n \mathbb{P}(N = n) && \text{(part (a), and } F_{Z_0}(z) = 1 = (F_{X_1}(z))^0) \\ &= \sum_{n=0}^{\infty} g_N(F_{X_1}(z)) \\ &= e^{m(F_{X_1}(z)-1)} && \text{(since } g_N(t) = e^{m(t-1)}) \\ &= e^{-me^{-z}}. \end{aligned}$$

Thus, we have that,

$$F_{Z_N}(z) = \begin{cases} 0 & \text{if } z < 0, \\ e^{-me^{-z}} & \text{if } z \geq 0. \end{cases}$$

Note that $\mathbb{P}(e^{-Z_N} < 0) = 0$. For $z \in (0, 1)$, we have that

$$\begin{aligned} F_{e^{-Z_N}}(z) &= \mathbb{P}(e^{-Z_N} \leq z) = \mathbb{P}(-Z_N \leq \log z) = \mathbb{P}(Z_N \geq \log(1/z)) \\ &= 1 - \mathbb{P}(Z_N < \log(1/z)) = 1 - F_{Z_N}(\log(1/z)) \\ &= 1 - e^{-me^{-\log(1/z)}} = 1 - e^{-me^{\log z}} = 1 - e^{-mz} \end{aligned}$$

Moreover, in the case that $z \geq 1$, we have that

$$F_{e^{-Z_N}}(z) = \mathbb{P}(e^{-Z_N} \leq z) = \mathbb{P}(-Z_N \leq \log z) = \mathbb{P}(Z_N \geq \log(1/z)) = 1,$$

since $\log(1/z) < 0$ and $\mathbb{P}(Z_N \geq 0) = 1$. Consequently, we have that

$$F_{e^{-Z_N}}(z) = \begin{cases} 0 & \text{if } z \leq 0, \\ 1 - e^{-mz} & \text{if } 0 < z < 1, \\ 1 & \text{if } z \geq 1. \end{cases}$$

Thus, $F_{e^{-Z_N}}(z) = F_W(z)$ for all $z \in \mathbb{R}$, and hence, $e^{-Z_N} \stackrel{d}{=} \min(1, Y)$.

Good luck!

SF2940: DISTRIBUTIONS AND THEIR CHARACTERISTIC FUNCTIONS

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a), a \in \mathbb{R}$	$p(a) = 1$	e^{ita}
Bernoulli	$\text{Be}(p); 0 \leq p \leq 1$	$p(0) = q, p(1) = p; q = 1 - p$	$q + pe^{it}$
Binomial	$\text{Bin}(n, p), n = 1, 2, \dots; 0 \leq p \leq 1$	$p(k) = \binom{n}{k} p^k q^{n-k}, k = 0, 1, 2, \dots, n; q = 1 - p$	$(q + pe^{it})^n$
Geometric	$\text{Ge}(p), 0 \leq p \leq 1$	$p(k) = pq^k, k = 0, 1, 2, \dots; q = 1 - p$	$\frac{p}{1 - qe^{it}}$
Poisson	$\text{Po}(\lambda), \lambda > 0$	$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$	$e^{\lambda(e^{it} - 1)}$

TABLE 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$U(a, b), a < b$	$f(x) = \frac{1}{b-a}, a < x < b$	$\frac{e^{itb} - e^{ita}}{it(b-a)}$
Exponential	$\text{Exp}(\lambda), \lambda > 0$	$f(x) = \frac{1}{\lambda} e^{-x/\lambda}, x > 0$	$\frac{1}{1 - \lambda it}$
Laplace	$L(\lambda), \lambda > 0$	$f(x) = \frac{1}{2\lambda} e^{- x /\lambda}, x \in \mathbb{R}$	$\frac{1}{1 + \lambda^2 t^2}$
Beta	$\beta(r, s), r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1} (1-x)^{s-1}, 0 < x < 1$	*
Normal	$N(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}, x \in \mathbb{R}$	$e^{i\mu t - \frac{1}{2}\sigma^2 t^2}$
Standard normal	$N(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, x \in \mathbb{R}$	$e^{-\frac{1}{2}t^2}$
Log-normal	$\text{LN}(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\log x - \mu)^2/\sigma^2}, x > 0$	*
Cauchy	$C(m, \eta), \eta > 0, m \in \mathbb{R}$	$f(x) = \frac{1}{\pi} \frac{\eta}{(x-m)^2 + \eta^2}, x \in \mathbb{R}$	$e^{imt - \eta t }$

TABLE 2. Continuous distributions